



Individual and team annotation effects on students' reading comprehension, critical thinking, and meta-cognitive skills[☆]

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ABSTRACT

Many students enter college without the needed skills to be successful. Colleges and universities are seeking instructional interventions to address these needs. Various classes are leveraging web-based social media to provide new instructional technologies that will help students learn. This paper reports on two studies related to the potential of online social annotation for improving teaching and learning in second-semester Freshman English classes. The approach, referred to as the Social Annotation Model-Learning System (SAM-LS), combines various instructional strategies, team-based learning, and a social annotation computer-supported collaborative learning tool, HyLighter, to increase student engagement with selected essays and with classmates. SAM-LS stimulates students to actively monitor their thoughts and compare them to both peers and the instructor (or domain experts). Study 1 showed no significant difference between the SAM-LS approach and a control; however, results appear to be related to confounding factors. Study 2 showed that students achieve better outcomes on measures of reading comprehension and meta-cognitive skill, but not critical thinking, when SAM-LS activities include small team collaborations. The two studies suggest future directions for research and development of SAM-LS and the HyLighter tool.

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1. Introduction

In order to learn, and later, to perform well in the work place, students need the skills to gain knowledge from written language. When it comes to academic success, reading, writing, and critical thinking are key skills that students need in order to be successful in the college classroom. The ability to read and ultimately learn from reading is a basic skill fundamental to scholastic success in any discipline of study (ACT, 2005). The acquisition and understanding of idea from text require various learning strategies, meta-cognitive skills for regulating the use of learning strategies, as well as the desire to use them.

Various studies have shown that high school graduates entering college lack essential skills for academic success. The American College Testing Program (ACT, 2005) has reported that only a quar-

ter of high school graduates were adequately prepared for college-level work in English, math, science, and reading comprehension. ACT also reported that only 51% of students who took their college entrance exam met college-readiness benchmarks in reading comprehension.

Regardless of the cause, students are coming to colleges and universities under-prepared. The three areas wherein students are lacking sufficient skills necessary for success in post-secondary education include reading comprehension, critical thinking, and meta-cognitive skills (ACT, 2005; Adelman, 1996; Hartman, 2001; Mendelman, 2007). Between 1970 and 2000, while college enrollments expanded at an annual rate of 1.98%, students' reading comprehension skills remained unchanged (Cox, Freisner, & Khayum, 2003). In a longitudinal study of the post-secondary educational experiences of under-prepared students, Adelman (1996) concluded that a deficiency in reading skills is a symptom of comprehensive literacy problems, which in turn significantly decreases the odds of a student completing any degree. The 2002 Condition of Education report suggested that a reading deficiency is the greatest barrier to under-prepared students' success in college (Wirt et al., 2002).

College freshmen need to develop and enhance their academic skills beyond their typically used passive learning strategies such

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as memorization and simple rehearsal (Anderson & Armbruster, 1984; Simpson & Nist, 1990). There is evidence that college students have a decreased tolerance for long reading assignments, and a declining ability to read critically and to extract meaning from what they read (Donahue, Voelkl, Campbell, & Mazzeo, 1999). Research shows that students perform better in post-secondary education when they have sufficient reading comprehension, critical thinking, and meta-cognitive skills (Cox et al., 2003; McCabe, 2000; Oudenhoven, 2002). Thus, if students are to be successful in college, it is vital that they gain these essential skills.

In response, colleges and universities are attempting to bridge this gap by seeking to identify and implement interventions to help students become more proficient as learners. The debate of how to help students improve these skills varies (ACT, 2005; Adelman, 1996; Boylan, 1999; McCabe, 2000; Oudenhoven, 2002). Universities spend resources, money, and time enrolling students in remedial training courses. Students take extra time to complete courses that are usually not counted toward graduation and incur extra expenses (Boylan, 1999; Oudenhoven, 2002). McCabe (2000) showed that annually more than one million US students enter college under-prepared and enroll in remedial courses (20% of the students in reading, 25% in writing, and 34% in math). It is evident that the challenge of helping prepare students for post-secondary education deserves more effort and attention, and most likely will require multiple solutions to deal with this comprehensive challenge. The common solution of offering remedial courses still leaves room for improvement and for other innovative solutions.

A possible alternative to remedial programs is to embed within students' regular first-year college curriculum training that focuses on the development of basic reading comprehension, critical thinking, and meta-cognitive skills. The training would require providing students with sufficient, effective, deliberate practice to help individual students progress along a trajectory from novice to proficient learner (Ericsson, 2002). This approach must be multi-faceted, incorporating the latest learning technologies to teach students about the skills they need to acquire, but of most importance, this approach needs to facilitate practice with feedback. Through appropriate interventions, under-prepared students may possibly achieve success in post-secondary education comparable to that of their adequately prepared peers (Boylan, 1999).

There are a number of ways to approach the challenge of designing a solution to address this skills-development need for many college students. For colleges and universities, the main vehicle for change would be to better structure courses so that there is better learning. Better learning strategies would make use of new technology and new pedagogical approaches. For our research, we have decided to focus on collaborative learning (learning in teams) as the key approach for enhancing student learning.

For this project, our main objective was to find a way by which students may gain essential skills for academic success through the appropriate implementation of social annotation technology. Specifically, we aimed at developing good pedagogical models using the social annotation tools with university students by investigating different instructional approaches. In this paper, we present the findings from two studies: Study 1 considered different instructional methods, whereas Study 2 evaluated the effect of working individually or collaboratively. Given these two settings, we studied student-learning outcomes based on the various experimental treatments.

1.1. Learning system framework

With increased access to computers and network technology, students and teachers are acquiring new means of dramatically improving teaching and learning techniques by leveraging recent

findings from different disciplines. Based on the thinking in several fields, the Social Annotation Model-Learning System (SAM-LS) was created with the intent to develop a learning model for development of student skills, specifically reading comprehension, critical thinking, and meta-cognitive skills. This intervention was created based on research from the areas of social annotation, instructional design, team-based learning, and computer-supported collaborative learning.

1.1.1. Social annotation

Using HyLighter,¹ SAM-LS not only shares information with the user, but also allows the user to highlight and annotate in an electronic medium (usually text). The use of HyLighter allows individuals to share their conceptual understanding of a text document with others.

Historically, a type of low-tech shared annotation process was widely practiced by medieval scholars who used the margins and spaces between lines of manuscripts to engage in dialogue with others. The same physical copy of a manuscript was passed around a community, and selected annotations (often including comments of readers on the remarks made by other readers) were customarily retained when scribes made new copies (Wahlstrom & Scruton, 1997; Wolfe, 2002). Today, social annotation systems allow students to engage in collaborative conversations in the margins of digital pages that are tied to specific sections of the pages. Integrating social annotation software into instructional materials seems an ideal approach for building on the social networking phenomenon. By making reading a shared experience, students may be more likely to engage with the text.

Today, social media are being used for educational purposes. Most wikis and blogs have provisions that enable contributors to have discussions either at the bottom of the page, below each paragraph, or through a separate tab, and to add links to other web pages or documents. In general, wikis and blogs lack a mechanism for engaging contributors in threaded discussions tied to specific sections of the page. Arguably, the provision of a highly granular discussion mechanism (i.e., the capacity for several or more readers to engage in threaded discussions tied to any section of a page without visually overwhelming the page or margins) has important implications for education.

Scientific evidence emphasizes the effectiveness of annotations (readers making comments in the margins of the text as they read) in increasing memory and student learning (Anderson & Armbruster, 1984; Bradley & Vetch, 2007; Glover, Xub, & Hardaker, 2007; Lavagnino, 1997; Porter-O'Donnell, 2004; Simpson & Nist, 1990). Clear evidence exists as to the effectiveness of annotations in increasing memory and learning of students (e.g., Anderson & Armbruster, 1984; Bradley & Vetch, 2007). In addition, proponents of active reading have advocated that students prioritize information by highlighting important points and adding comments (Adler & Van Doren, 1972; Brown & Campione, 1990). On the other hand, research on the effects of highlighting (and underlining) as a study technique provides evidence that many students are not skilled at identifying what is relevant in a text. Several studies have found that students in high school and the freshmen level of college generally make poor choices in deciding what information to highlight or underline (Peterson, 1992; Schellings & van Hout-Wolters, 1995).

Lavagnino (1997) argued that marginal annotation both clarifies and fixes the initial reaction to the text, and thereby, can contribute to the person's process of interpretation. Bradley and Vetch (2007) have even suggested that the digital age has not produced the expected transformation in scholarship because the capacity for end-users to annotate digital pages is not widely available. As a result, readers tend to be passive and not deeply engaged.

¹ www.hylighter.com.

Annotation systems are playing an increasing role in the development and dissemination of scientific knowledge and data. This trend is helping to transform the machinery of knowledge creation itself by enabling scientists on an unprecedented scale to participate in globally distributed knowledge-building efforts (Buneman & Steedman, 2002). However, it is still unclear how best to use social annotation systems to help facilitate learning and improve the quality of instruction.

1.1.2. Instructional design

SAM-LS's basic learning task involves having each student read, think, write, and reflect. The use of social annotation software may increase students' learning by providing appropriate instructional strategies. However, if learning activities are not aligned with an effective instructional strategy, students are likely to lose motivation and interest (van Merriënboer, 1997).

Merrill (2002) has proposed a four-phase instructional-design model that provides a practical approach to integrating motivational and instructional strategies to promote efficient, effective, and engaging learning. This approach, *First Principles of Instruction*, guided the current research effort. Principles include (a) the task-centered principle (e.g., promote learning by embedding instruction in the context of whole, real-world tasks), (b) the activation principle (e.g., help learners activate relevant cognitive structures by directing them to recall, describe, or demonstrate relevant prior knowledge), (c) the demonstration principle (e.g., promote learning by providing demonstrations of skills to be learned that are consistent with the type of content being taught), (d) the application principle (e.g., promote learning by engaging students in application of their newly acquired knowledge or skill through activities that are consistent with the type of content being taught), and (e) the integration principle (e.g., promote learning by directing students to integrate their new knowledge into their everyday life through reflecting-on, discussing, or defending their new knowledge or skill).

1.1.3. Team-based learning

In SAM-LS, multiple learners may interact with each other in sharing their thoughts and responding to one another. These types of activities may require the students to engage in critical thinking. One area of research that provided support for the design of SAM-LS was team-based learning (TBL). Many educators are moving away from the typical lecture-style teaching, and onto TBL, which is more interactive and engaging (Lightner, Bober, & Willi, 2007). Team-based learning is an instructional method that requires students to work together on task that go beyond memorization and acquisition of knowledge. Three key principles guide the implementation of this strategy, which include hold students accountable for individual learning, provide learning tasks that require students to apply prior knowledge, assignments should promote team development as well as individual and team learning, and teams need to receive immediate feedback.

The integration of team-based learning has proven an effective learning strategy in various studies. These studies have shown a significant increase in learning and performance outcomes when students engaged in team-based learning practices (Clark, Nguyen, Bray, & Levine, 2008; Haberyan, 2007; Thompson et al., 2007). Not only does this method seek to increase individual comprehension and skill development, but it also seeks to increase student engagement (Michaelsen, Knight, & Fink, 2004). Michaelsen also points out that when a functional team joins intellectual power, they synergize one another's intelligence, and end up creating more than if they engaged in the learning individually. The benefits from the TBL method include active engagement in problem-solving, group collaboration, learning how to deal with other people, leadership skills, self-esteem, awareness of the diversity of settings, deeper

learning, and better retention of course content (Lancaster & Strand, 2001). While there are benefits to learning in teams, there are some challenges. Most of the problems according to Michaelsen, such as conflict and free riders, can be addressed by following the principles of TBL.

1.1.4. Computer-supported collaborative learning

Another line of research that had implications on the design of SAM-LS is the area of computer-supported collaborative learning (CSCL). CSCL states that information and communication technology in education more or less explicitly considers technology's possibilities to facilitate social interaction (Koschmann, Hall, & Miyake, 2002; Kumpulainen & Wray, 2002). The purpose of CSCL is to support students in effective collaborative learning. The features of the technology are meant to facilitate students in managing their collaboration goals including any requirements of a particular collaborative task (Lehtinen, 2003), and more generally to promote reflection and inquiry that assist the in-depth learning (Dillenbourg, 2002). CSCL systems facilitate communicating ideas and information, accessing information and documents, and providing feedback on various learning activities.

CSCL has several advantages for team learning, especially when the team is working on complex tasks or problems. Researchers (Gokhale, 1995; Johnson & Johnson, 1991; Lehtinen, 2003; Srinivas, 2008) have presented several benefits of CSCL, which may help (a) develop skill building and practice, (b) enhance self management skills, (c) develop oral and written communication and social interaction skills, (d) increase the sharing of ideas and shared understanding, (e) establish a sense of learning community, (f) increase student motivation, (g) encourage different perspectives, (h) develop higher level, critical-thinking skills, (i) aid in the development of meta-cognitive and evaluative thinking skills, and (j) facilitate deeper learning.

In order to support learning, the CSCL environment must attend to both technical and the social aspects of collaboration (Kirschner, Strijbos, Kreijns, & Beers, 2004). Many of the technical aspects are dependent on the types of interaction functions that are needed in order to collaborate. However, there are social aspects that go beyond the functionality of the CSCL tool and into the team interaction design. Of particular interest are two basic principles that when followed provide an ideal implementation of CSCL. These are: (1) interaction facilitation, and (2) conflict generation (Collazos, Guerra, & Pino, 2004). One of the most important aspects of collaborative learning is to facilitate the students' learning by monitoring the teams, and intervening to provide task assistance or to increase students' interpersonal and team skills. At times, facilitators should provide assistance when needed in order for the students to complete the task accurately and effectively. Consider the second principle, teams may become stronger as they work through conflicts that arise within the collaboration. When two individuals disagree, at some point they feel a social pressure to solve that conflict, and the resolution of this conflict may lead one or both of them to change their viewpoint. Individual learning is not initiated and generated by the conflict itself, but by its resolution; that is, by the justifications and explanations that lead to a jointly accepted proposition. No matter how well a facilitator prepares, conflicts naturally occur and as such, facilitators need to keep in touch with the team progress through periodic progress reports or team assessments. Such a monitoring process allows the instructor to identify problems within the teams as they arise.

1.2. Annotation technology

Using HyLighter as the online annotation tool, SAM-LS was designed based on the combination of social annotation, instructional design, team-based learning, and CSCL frameworks. The two stud-

ies reported here employed HyLighter as the technology tool to create the learning system (SAM-LS). Preliminary pilot tests on social annotation activities at a community college identified the following potential benefits: (a) increased participation, engagement, accountability, and completion rates in local and distance learning modes, (b) developed active reading skills, enhanced learning from text and improved writing skills, (c) improved quality of instruction, (d) helped identify specific obstacles to learning, and (e) improved assessment of complex and higher-order cognitive skills (Lebow, Lick, & Hartman, 2004). These initial findings suggest that social annotation activities have the potential to facilitate comprehension, memory, and transfer as students learn from and with each other and their instructor.

1.3. Social annotation and learning outcomes

Based on the pilot studies, it appeared that a social annotation strategy has the potential to have a positive effect on a variety of learning outcomes. With these potential effects, would a systematic implementation of an individual and social annotation strategy have a broader impact across a variety of skills sets? Specifically, the researchers aimed at determining if an annotation strategy had an effect on lower-level, moderate-level, and complex-level cognitive skills. The skill levels were operationalized to work within a typical second-semester Freshman reading and writing class. The specific skills ranged from reading comprehension skills, to critical-thinking skills, and to meta-cognitive skills.

The acquisition of reading comprehension skills is vital to the success of college students. Mendelman (2007) argued that students go through the steps of critical thinking through highly structured exercises, but avoid practicing critical thinking. As a result, they do not learn how to think critically. According to Mendelman, the practice of critical thinking is somewhat apparent in high school curriculum, but not necessarily taught as a skill to students. Students may be going through the motions of critical thinking, but may not actually understand what they are doing.

In a reading context, if students are asked to provide an interpretive thesis statement, they are required to engage in creative and outside-of-the-box prescribed thinking (Mendelman, 2007). This activates Bloom's higher-order thinking skills of analysis, synthesis, and evaluation (Krathwohl, 2002). Long-term increase in critical thinking requires engagement in critical thinking exercises for extended periods of time. Furthermore, unless students are actively engaged in the practice of critical thinking, these skills are likely to remain underdeveloped.

Meta-cognition, or the act of thinking about one's own thinking (Hartman, 2001), is an equally important academic skill for students to learn. Similarly to critical thinking, development occurs when students engage in the practice of meta-cognition. Meta-cognition involves two basic types of "thinking about thinking." One type includes executive management or control strategies (the "boss skills") for planning, monitoring, evaluating, and revising one's thinking processes and products. The other includes strategic knowledge about what information and strategies/skills one has available (i.e., declarative knowledge), when and why to apply them (i.e., contextual or conditional knowledge), and how to use them and evaluate their effectiveness (i.e., procedural knowledge (Hartman, 2001; Sternberg, 1986).

While each of these skills is conceptually different, the development of one's skills enables the development of another skill. Ever-son and Tobias (1998) showed that students with the ability to determine their level of understanding in a certain subject area were better able to identify their weaknesses in specific areas of math, and were also better able to learn previously lacking skills, predict their future scores, and achieve higher academic performance. Higher academic performance resulted from students iden-

tifying their areas of weakness, focusing on improving those areas, and ultimately, wasting little time on things that they already understood. Thus, it is evident how meta-cognitive skills help support lower-level learning. As such, the acquisition of meta-cognitive skills seems vital to the success of students progressing through undergraduate programs.

The research presented herein consists of two studies. The first study examined the effect of different social annotation practices enabled by HyLighter on the reading comprehension, critical thinking, and meta-cognitive skills of students in higher education settings. A design that included five different treatments, or practices, was developed. Students were asked to (a) highlight and annotate selected articles, (b) highlight and annotate selected articles and review peer highlights and annotations, (c) highlight and annotate selected articles and review instructor's highlights and annotations, (d) highlight and annotate selected articles after previewing instructor's highlights and annotations and performing a self-reflection task (i.e., compare their own highlights and annotations to the instructor's highlights and annotations but with no review of peer input), and (e) read a hard copy of a selected article (i.e., without HyLighter and a specified practice).

Study 2 was designed to examine the effect of social annotation practices on the same cognitive skills as in Study 1 (i.e., reading comprehension, critical thinking, and meta-cognitive skills), but the focus was on differences in outcomes between group and individual activities. Four instructional methods were devised involving a two-step procedure. In step one, students completed HyLighter activities (e.g., highlight and comment on articles and respond to questions) either collaboratively in small groups or individually. In the second step, students compared their responses to the instructor's highlighting and comments either collaboratively in small groups or individually. This design is set up to create four different conditions and examine their effect on learning outcomes.

2. Method of Study 1

2.1. Sampling

2.1.1. Students

Two hundred and fifty-four students (M age = 20.71 years, SD = 4.7) enrolled in a southeastern community college second-semester Freshman English course that focused on argument and persuasion participated in this study. Of these participants, 43% were Female, 39% were Male, and 19% were unspecified; 30% were Black, 41% Caucasian, 2% Asian, 0.3% Native, 6% Hispanic, and 6% were unclassified.

2.1.2. Instructors

The sampling is comprised of students from nine sections. Five different instructors taught these nine sections. One instructor, who taught just one section, was the coordinator for the course sections, and a full-time faculty member of the community college. The other four instructors, teaching two sections each, were adjunct faculty. The course enrollment was: section 1 (n = 31), section 2 (n = 25), section 3 (n = 30), section 4 (n = 30), section 5 (n = 26), section 6 (n = 29), section 7 (n = 29), section 8 (n = 28), and section 9 (n = 26).

2.2. Instrumentation

The instruments were designed to measure three cognitive skills applied to five articles. Two additional articles served as pre- and post-tests. The developmental procedures of the three skills instruments were similar. The questions for each of the

instruments were written by the coordinator and reviewed and edited by an evaluation team of three experts in the field. The final version of each instrument's questions consisted of items that the coordinator and the evaluation team agreed upon. Although the questions for each instrument were specific to the corresponding article, all questions were relevant to reading comprehension, critical thinking, and meta-cognitive skills.

2.2.1. Reading comprehension skills instrument (RCSI)

The RCSI comprises three questions aimed at assessing reading comprehension skills. Two sample reading comprehension questions are: "What, according to Schiffren, were the negative results of the "sexual revolution?" and "According to Reed, what would be a major strategy that the courts use to identify possible gender discrimination?" Responses were rated as either correct or incorrect for each question. The overall score for the RCSI was comprised of the number of correct responses to all the questions divided by 3.

2.2.2. Critical-thinking skills instrument (CTSI)

The CTSI comprises two questions aimed at assessing critical-thinking skills. Two sample critical-thinking questions are: "Although most people would acknowledge that gender discrimination may play a role in career advancement, what other factors could contribute to this workplace phenomenon?" and "What is Givhan's thesis?" The research team categorized responses as 0 (*did not attempt, missed the point of the questions*), 0.5 (*rational, but incoherent response*), and 1 (*rational, coherent response*). Summarizing the total responses and dividing by 2 comprised the final score of the CTSI.

2.2.3. Meta-cognitive skills instrument (MCSI)

The MCSI comprises two questions aimed at assessing meta-cognitive skills. Two sample meta-cognitive questions are: "Explain the rationale or reasons for your conclusion," and "After

answering the above questions, write down the sequence of steps you went through in your mind in forming your response. Describe in detail the actual thoughts you were having at each step." The research team categorized responses as 0 (*did not attempt, missed the point of the questions*), 0.5 (*rational, but incoherent response*), and 1 (*rational, coherent response*). Summarizing the total responses and dividing by 2 comprised the final score of the MCSI.

2.3. Instructional tool

This study used HyLighter as the online annotation tool. In general, annotating documents online is comparable to paper and pencil annotations in which readers highlight text and make comments. Contributors use the cursor to block the text upon which they wish to comment, and a comment box opens for their input. What makes HyLighter different are its mechanisms for: (a) aligning commentary with related sections of the document being discussed, despite the limited real estate available in the margins; and (b) enabling virtually any number of contributors to add highlighting and comments to web pages. In brief, HyLighter "maps" the intellectual travels of any size group by adding a layer of color-coded highlighting on top of a web page linked to comments in the margins. This technique enables a few, several, or a large number of people to engage in collaborative conversations tied to specific sections of a document without visually trampling the primary text or cluttering the margin.

As group members annotate a document, HyLighter uses color-coding to map the distribution of inputs from the group (Fig. 1). An area highlighted by you (the logged-in user) but not by anyone else appears in yellow; areas not highlighted by you, but marked by one or more contributors, appear in shades of blue (the darker the shade, the more overlapping interest for that fragment); and areas highlighted by you and others appear in various shades of green. Comments associated with each highlighted area are listed in the margin or comment field. In addition, HyLighter provides a variety

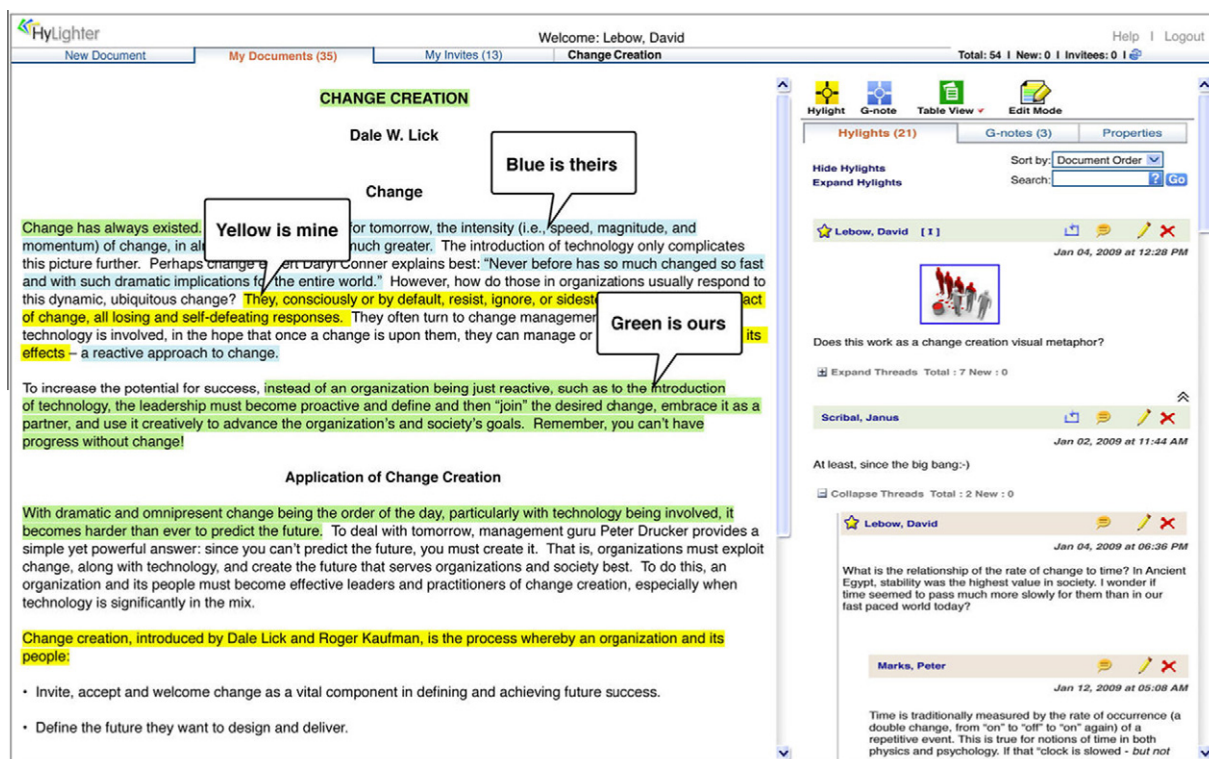


Fig. 1. Color-coded distribution map including comments in the margin.

of “views” for reviewing the group input, including a table showing highlighted excerpts and associated comments with various search and sort options (e.g., by username, recommended changes, or date modified).

2.4. Instructional methods

The instructional methods included various instructional activities using the HyLighter software. Four of the five instructional methods incorporated HyLighter but used different combinations of its features. The fifth instructional method is the control condition (Table 1). The rationale for this approach was to measure the effects of highlighting and annotation alone and in combination with three different practices on student learning (reading comprehension, critical thinking, and meta-cognition). The instructional variations are outlined in Table 1. For each instructional method, the students read an article and then answered a series of questions that comprised the RCSI, CTISI, and MCSI.

In instructional method 1, students read an article presented in HyLighter and were encouraged to highlight and annotate the article as they felt appropriate to help them better understand the text. Students highlighted and annotated elements individually and did not see the highlights or annotations of any other student. No other prompts or instructions were provided.

For instructional method 2, students completed the same tasks given in instructional method 1 followed by each student viewing the highlights and annotations of their peers.

Instructional method 3 involved students completing the same tasks given in instructional method 1 followed by each student viewing the highlights and annotations of the lead instructor.

Instructional method 4 required the students to review the lead instructor's highlights and annotations prior to reading the article. Students referred to the lead instructor's highlights and annotations as they completed the same tasks given in instructional method 1. After submitting their own highlights and comments, students answered two questions: (a) “What is similar or different between your annotations and the lead instructor's annotations?” and, after analyzing the lead instructor's annotations, (b) “What changes would you make to your annotations?” Of note, the rationale for allowing students to preview the instructor's highlights

and annotations was to examine the effects of providing an expert mental model as a scaffold of the desired outcomes. The reasoning was that previewing the model would help students perform better on the highlight and annotate activity and, as a consequence, would lead to improved learning outcomes.

Instructional method 5 (control condition) consisted of students reading a printed article from a textbook without using HyLighter. No other prompts or instructions were provided for this treatment condition.

2.5. Procedures

Prior to the implementation of the five instructional methods (including the control), the five instructors for the course's nine sections were introduced to HyLighter and all of its technical features. The research team met with the instructors for 6 h, and all of the software instructions were reviewed, as were the different instructional methods and the study's procedures.

The articles included in the study were selected by the lead instructor, and reviewed by the research team and the other instructors. Once the article selection was finalized, the instructors carefully read each article and then wrote questions corresponding to RCSI, CTISI, and MCSI.

Following training and instrument development, the research team began the study. Each course section followed the same sequence of events, including the same sequence of articles, except that the order of the instructional methods varied in accordance with a Latin square design (Table 2).

Prior to instructional methods implementation, students within each section were given a demographic questionnaire and a pre-test that included an article that was not taught by any of the instructional methods. Students were asked to read the pre-test article (print-based) carefully, and complete the reading comprehension, critical thinking, and meta-cognition skills instruments (RCSI, CTISI, and MCSI, respectively).

Once the pre-test had been completed, each course section began the instructional activities starting with article 1 and continuing until all five articles were covered over a period of 10 weeks. Each section used a specific instructional method depending on the specific article (Table 1). All five instructors used each of the five instructional methods over the course of the study. The research team led each of the sessions of the study while the instructors were present. This procedure secured equal instructions applied to each article across all instructors. After each session, each student completed the skills instruments (RCSI, CTISI, and MCSI) pertaining to the specific article.

After the instructional activities had been completed, a post-test on an additional article was conducted. Students were asked to read the post-test article (print-based) carefully and complete the reading's skills instruments (RCSI, CTISI, and MCSI). Two weeks after all the data were collected, the instructors were invited to a research team meeting where they were debriefed and given the

Table 1
Instructional methods used in Study 1.

Method	Description
1	Highlight and annotate (H&A)
2	H&A + review of peer highlights and annotations
3	H&A + review of lead instructor highlights and annotations
4	Preview lead instructor highlights + H&A + self-reflection (i.e., review and compare to expert highlights and annotations but no review of peer input)
5	Read hard copy of article, no use of HyLighter (control)

Table 2
Latin square repeated measures design used in Study 1.

Instructor/course section	Pre-test	Intro to HyLighter	Article 1	Article 2	Article 3	Article 4	Article 5	Post-test
1/a	Pre-test	Booklet	1	2	3	4	5	Post-test
1/b	Pre-test	Booklet	1	2	3	4	5	Post-test
2/a	Pre-test	Booklet	2	3	4	5	1	Post-test
2/b	Pre-test	Booklet	2	3	4	5	1	Post-test
3/a	Pre-test	Booklet	3	4	5	1	2	Post-test
3/b	Pre-test	Booklet	3	4	5	1	2	Post-test
4/a	Pre-test	Booklet	4	5	1	2	3	Post-test
5/a	Pre-test	Booklet	5	1	2	3	4	Post-test
5/b	Pre-test	Booklet	5	1	2	3	4	Post-test

opportunity to share their experiences and reflections about HyLighter and the instructional methods.

3. Results of Study 1

3.1. Task-independent cognitive components

Repeated measures (RM) multiple analysis of variance (MANOVA) using time (pre-post interventions), and three cognitive components as repeated measures resulted in significant time effect, $Wilks' \lambda = .95$, $F(1, 118) = 5.77$, $p = .02$, $\eta_p^2 = .05$. On average, students exhibited slightly better cognitive skills after exposure to the learning methods (Effect Size; $ES = .26$). Also, the cognitive components resulted in significant effect, $Wilks' \lambda = .38$, $F(1, 117) = 96.24$, $p = .000$, $\eta_p^2 = .62$. On average, comprehension was higher than critical thinking ($ES = .97$), and both more pronounced than meta-cognitive skills ($ES = 2.01$, and $ES = 1.04$, respectively). Finally, the time by cognitive skill resulted in a non-significant but meaningful effect, $Wilks' \lambda = .96$, $F(2, 117) = 2.52$, $p = .08$, $\eta_p^2 = .04$. This effect is displayed in Fig. 2. Students did not exhibit any change in meta-cognition ($ES = 0.0$), nor in comprehension ($ES = .12$). Increase was noticed only in critical-thinking skills ($ES = .43$).

3.2. Task-related cognitive components

Repeated measures MANOVA using cognitive components (i.e., comprehension, critical thinking, and meta-cognition) and instructional methods as repeated measures factors were employed. Neither instructional method nor its interaction with cognitive components resulted in significant effects ($p > .05$). The main effect of cognitive component was significant, $Wilks' \lambda = .28$, $F(2, 51) = 63.88$, $p = .000$, $\eta_p^2 = .72$. On average students exhibited more critical-thinking skills than comprehension skills ($ES = 1.42$) and meta-cognition ($ES = 2.75$), and more comprehension skills than meta-cognition ($ES = 1.33$). Differences in the three cognitive components are illustrated in Fig. 3. The non-significant instructional method by cognitive component effect is depicted in Fig. 4. Meta-cognition was the least acquired skill under all instructional methods, and critical thinking was the highest, but similarly under the five instructional methods.

4. Discussion of Study 1

The main purpose of Study 1 was to measure the effects of four different highlighting and annotation practices on reading compre-

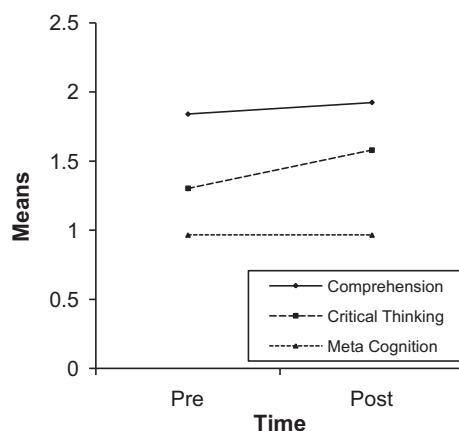


Fig. 2. Means for three cognitive components (reading comprehension, critical thinking, and meta-cognition) prior to and after completion of five article activities across instructional methods.

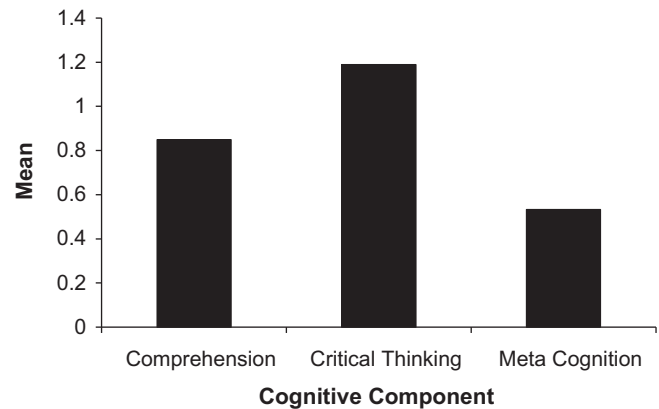


Fig. 3. Mean values for the three cognitive components (reading comprehension, critical thinking, and meta-cognition) across instructional methods.

hension, critical thinking, and meta-cognition. The findings revealed no significant difference on the three outcome variables between the SAM-LS instructional interventions and the instruction omitting the SAM-LS, indicating insufficient evidence for the notion that SAM-LS results in superior learning outcomes. However, the SAM-LS instructional methods did not have a detrimental effect on students learning either.

In reflecting on these findings, one can identify two confounding effects, which could impede the SAM-LS instructional effect: (1) limited exposure time to the intervention, and (2) limited time to learn how to use HyLighter. Due to the complexity of learning a new cognitive tool and getting up to speed, the treatment conditions perhaps did not engage students long and deeply enough in the text and with each other in order to show a significant difference (Moss, 2000). In order to obtain a learning effect, the students may need more time to practice than what was provided in the intervention conditions, and a more prolonged period of time to obtain an overall net learning gain from using the SAM-LS interventions.

Study 1 pre- and post-test means across all treatments showed improvement in critical thinking, but not in the other two outcome measures. Due to the differences in the outcome variables complexity, one might expect to see differences from pre- to post-test measures for reading comprehension, but perhaps no differences

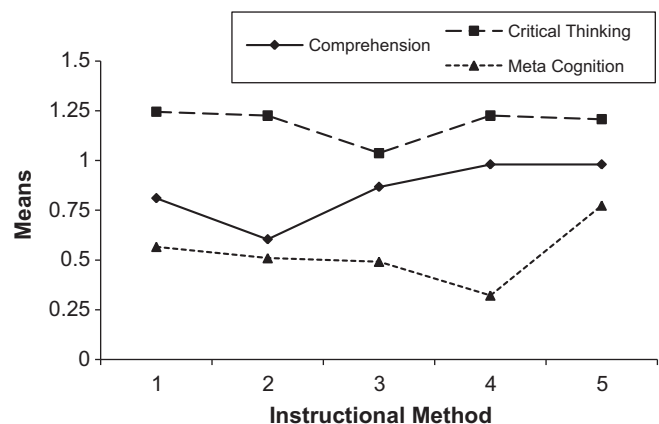


Fig. 4. Means for three cognitive components (reading comprehension, critical thinking, and meta-cognition) under four instructional methods (1 – highlight and annotate (H&A); 2 – H&A + review of peer highlights and annotations; 3 – H&A + review of lead instructor highlights and annotations; and 4 – preview lead instructor highlights + H&A + self-reflection) and one control (5 – read hard copy of article, no use of HyLighter).

for meta-cognition (Krathwohl, 2002); however, this was not the case. While there was not a significant difference across all treatment conditions and the control, this change on the critical thinking measure does represent a change for all the students in all instructional conditions including the control, which supports the idea that there was an overall benefit for supporting the learning of desirable critical-thinking skills for the specific type of English course.

5. Method of Study 2

5.1. Sampling

5.1.1. Students

Study 2 comprised 267 students (M age = 19.98 years, $SD = 3.2$) enrolled in a southeastern community college second-semester Freshman English course that focused on argument and persuasion. The sampling comprised students enrolled in 10 sections. Participants comprised 42% Female, 42% Male, and 17% unspecified; 17% Black, 27% Caucasian, 28% Asian, 5% Native, 0.4% Middle Eastern, and 2% unspecified.

5.1.2. Instructors

Five different instructors taught each of the 10 sections. The lead instructor was the coordinator for all of the course sections, and a full-time faculty member of the community college. The other four instructors were adjunct faculty who taught two sections each.

5.2. Instrumentation

The instrumentation for Study 2 was identical to that of Study 1 (see instrumentation section for Study 1), but the questions pertained to the different articles used for this study.

5.3. Instructional tool

The instructional tool for Study 2 was identical to that of Study 1 (see instructional tool section for Study 1).

5.4. Instructional methods

The instructional methods for Study 2 were assigned to each course section for the duration of the study. Specifically, each course section used the same HyLighter method (i.e., method 1, 2, 3, or 4) on all five articles. The variations are outlined in Table 3. For each instructional method, the students read an article from the same set of articles and in the same order for all instructional methods, completed the tasks associated with the instructional method, and then answered a series of questions that comprised the RSCI, CTSI, and MCSI.

The four instructional methods (Table 3) included the following: GA–GC—groups of students collaboratively complete the HyLighter activities, and then as the same group composition, collaboratively compare their responses to the responses from the lead instructor. IA–GC—individual students complete the HyLight-

er activities, and then put into a newly formed group collaboratively compare their responses to the responses from the lead instructor. GA–IC—groups of students complete the HyLighter activities, and then individually compare their group responses to the responses from the lead instructor. IA–IC—individual students complete the HyLighter activities, and then compare their individual responses to the responses from the lead instructor (IA–IC).

For each HyLighter activity, the students worked in either individual or group mode. First, they read an article presented in HyLighter and were then encouraged to highlight and annotate the article, as they felt appropriate to help better understand the text. After completing this task, they answered reading comprehension questions, “chunked” the article (i.e., performed the process of breaking down a text into smaller pieces in order to make sense in relation to the big picture), and identified the article’s thesis (i.e., identified a section of the article as a statement of the author’s thesis by highlighting and annotating the section). For the model comparison, students compared their responses to those of the lead instructor either individually or in small groups. Finally, after completing the HyLighter activity and model comparison for an article, students answered a series of questions that comprised the RSCI, CTSI, and MCSI.

5.5. Procedures

The procedures prior to the implementation of the four instructional methods were similar to those in Study 1, including instructor training, review of the HyLighter functions, review of the instructional methods and activities, and creating and reviewing the RSCI, CTSI, and MCSI. The implementation of Study 2 included an introduction to HyLighter prior to the five instructional activities. Each course section used the same articles in the same order.

6. Results of Study 2

Unlike Study 1, all the articles were taught using different instructional methods, thus being a clear between-subjects (BS) design with the five learning tasks as repeated measures. In this study the summative achievement tests consisted of one reading comprehension test, three critical thinking tests, and one meta-cognition test. Each was subjected separately to RM MANOVA. Because significant ($p < .05$) sphericity was obtained, the Greenhouse–Geisser (GG_{ms}) procedure replaced the commonly used Wilks’ λ multivariate estimate, and the df have been adjusted accordingly. The MANOVAs results are depicted in Table 4.

The findings resulted in several significant effects. First, the five learning tasks differed significantly ($p < .05$) across the instructional methods. These effects were expected due to inherent variance in difficulty level of the reading tasks selected for learning. Instructional methods, however, resulted in significant main effects for reading comprehension ($p = .03$), and a strong tendency ($p = .06$) for meta-cognition. These effects are illustrated in Fig. 5.

Students in the IA–IC instructional method exhibited lower meta-cognitive skills than students in the GA–GC ($ES = -.75$), IA–IC ($ES = -.78$), and GA–IC ($ES = -.76$). The differences among the last three instructional methods were negligible and non-significant ($p > .05$). Similarly, students in the IA–IC instructional method exhibited lower reading comprehension skills than students in the GA–GC instructional methods ($ES = -1.01$), IA–GC ($ES = -.87$), and GA–IC ($ES = -.38$). Students in the GA–GC scored higher than students in the GA–IC ($ES = -.49$).

Finally, the instructional method by learning task interaction analyses revealed a significant ($p < .05$) interaction for reading

Table 3
Instructional method used in Study 2.

Method	HyLighter activity	Model comparison
1	Group activity (GA)	Group comparison (GC)
2	Individual activity (IA)	Group comparison (GC)
3	Group activity (GA)	Individual comparison (IC)
4	Individual activity (IA)	Individual comparison (IC)

Table 4

RM MANOVA results for five cognitive outcomes by learning task and instructional method (IM).

Variable	Effect	GG_{ms}	F	df	p	η_p^2
Reading comp.						
A. Task		1.17	23.77	3.2, 163.20	.00	.32
B. IM			3.28	3, 51	.03	.16
C. A by B		.174	3.54	9.6, 163.20	.00	.22
Meta-cognition						
A. Task		7.38	12.74	2.92, 146.18	.00	.20
B. IM			2.632	3, 50	.06	.14
C. A by B		1.57	2.71	8.77, 146.18	.01	.14
Critical thinking 2						
A. Task		7.01	1.68	3.5, 154.05	.17	.04
B. IM			1.10	3, 44	.36	.07
C. A by B		7.35	1.76	10.5, 154.05	.07	.11
Critical thinking 3						
A. Task		.70	5.72	3.08, 157.11	.00	.09
B. IM			1.185	3, 51	.32	.07
C. A by B		.27	2.01	9.24, 157.11	.04	.11
Critical thinking 1						
	Wilks' λ	F	df	p	η_p^2	
A. Task		.83	1.99	4, 40	.11	.17
B. IM			.69	3, 43	.57	.05
C. A by B		.65	1.55	12, 106.12	.12	.13

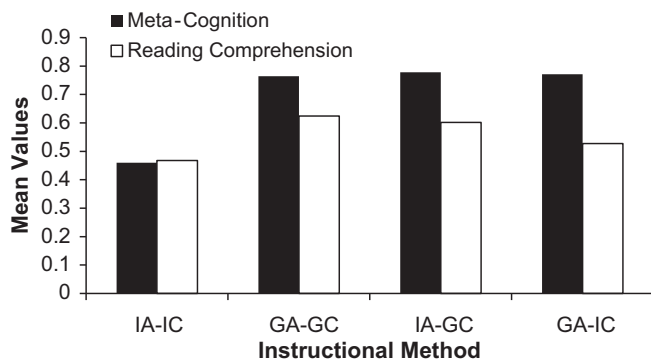


Fig. 5. Means for meta-cognition and reading comprehension for each of the four instructional methods (individual activity (IA)–individual comparison (IC); group activity (GA)–group comparison (GC); individual activity (IA)–group comparison (GC); and group activity (GA)–individual comparison (IC)).

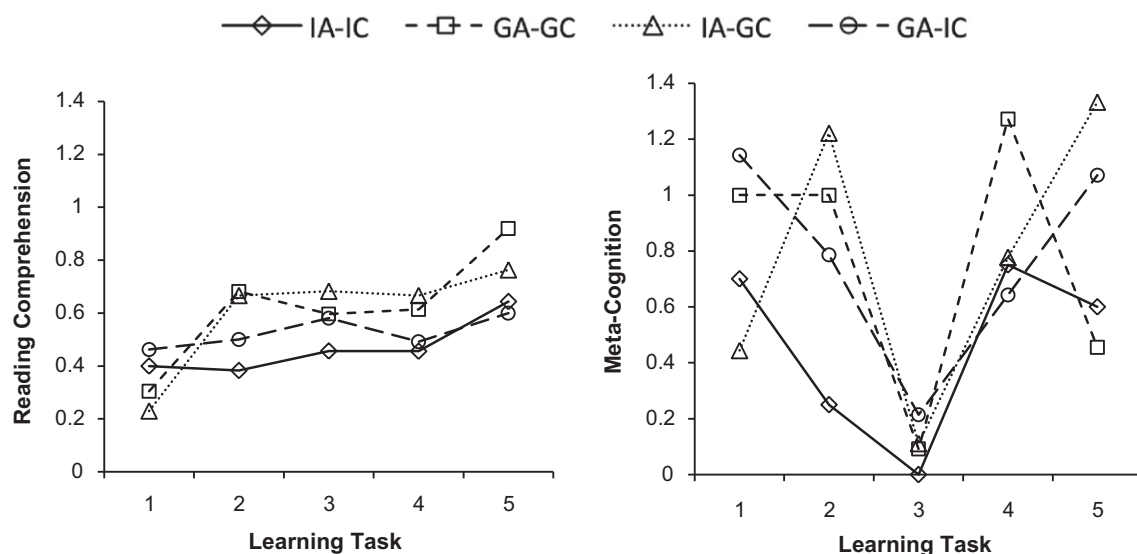


Fig. 6. Means for reading comprehension (left) and meta-cognition (right) for each learning task and the four instructional methods (individual activity (IA)–individual comparison (IC); group activity (GA)–group comparison (GC); individual activity (IA)–group comparison (GC); and group activity (GA)–individual comparison (IC)).

comprehension and meta-cognition but not for the critical thinking components. The two-way interactions are presented in Fig. 6.

The means for reading comprehension indicate that students exposed to GA–GC benefited most (e.g., outsourced students in other instructional methods) in the hardest learning task (learning task 5), but not in the easiest learning task (learning task 1). Also, students exposed to IA–GC benefited more in learning tasks 2, 3, and 4, but not in learning task 1. Students in IA–IC were behind in most learning tasks but learning task 1.

The means for the meta-cognitive skill indicate that after learning task 3, students in all learning methods were unable to demonstrate any meta-cognitive skills. However, following learning task 2, the ones who benefited most were those in the GA–GC and IA–GC. Students in GA–GC were also better than the students in the other instructional methods on learning task 4, while students in IA–GC and GA–IC were the best critical thinkers following learning task 5.

7. Discussion of Study 2

Results of Study 2 show that students benefited in reading comprehension and meta-cognition when they completed either one or both of the tasks (annotation and/or comparison) in groups. No such effect appeared for critical-thinking skills.

When looking at the effect of the treatments over time, one finds an interesting effect in task 3. The uniformly low scores on meta-cognitive skills for task 3 suggest that the task article was more demanding, thereby perhaps leaving fewer resources for meta-cognitive processing (Hoffman, McCurdden, Schraw, & Hartley, 2008). Since the task requirements were similar to the previous two and the following two tasks, it is unlikely that this was caused by a lack of understanding the task requirements. Nonetheless, as mentioned, the instructional method main effect is statistically significant across all five tasks.

When students engaged with the text in a collaborative mode by either highlighting and annotating the text and/or comparing their responses to the instructor's or a benchmark, learning outcomes for reading comprehension and meta-cognition improved, but there is a lack of evidence to suggest that there is a difference for critical thinking. In the group condition there appears to be an effect that is enhancing reading comprehension and meta-cognition. This effect could be explained by a resolution

of a conflict (Collazos et al., 2004). This study did not attempt to measure conflicts nor resolutions. Given the two parts to the instructional activity, a conflict and subsequent resolution would happen either while the students were interacting among themselves, or as each student team examined the instructors' annotations. In either case, where a team was present, there was no statistical difference in the students' reading comprehension nor meta-cognitive skills.

The basic task of highlighting and annotating focuses on reading comprehension. Any differences (conflicts) would be easily seen and therefore would be directly targeted for resolution. It is not so clear as to how conflicts would impact meta-cognitive skill development, but one could imagine how just as a conflict caused by the team would promote resolutions, this act could put each team member in a more reflective state whereby students begin to thinking about their own thinking (meta-cognition practice) resulting in strengthening their own awareness of how and what they think.

Regarding critical-thinking skills, students might not have the time nor cognitive resources to invest in pursuing more critical discussion about the reading. As a result, conflicts might not have been present to the team therefore substantively diminishing or even eliminating a CSCL effect of challenge the team members to engage in critical thinking. It might be possible that the learning activities in which the students were engaged were not specifically developing critical-thinking skills (Fisher, 2001; Lockhead & Clement, 1979). This finding raises additional questions: are collaborative modes of learning more effective for reading comprehension and meta-cognitive skills than for critical thinking? How might such embedded approaches affect basic reading comprehension, critical thinking, and meta-cognitive skills training if practiced over longer periods of time? For example, if students improve in their performance of the highlight and annotate task, and the compare task, would we see comparable improvements in learning outcomes?

Results from Study 2 suggest that the interventions had only modest effects on learning. However, results do show that when students engaged in the process of comparing their responses to a benchmark (i.e., the instructor's response), they benefited from collaborative effort in line with Collins, Brown, and Newman's (1989) findings. Of greater interest is the observation that students exposed to the instructional methods incorporating a group component scored higher than the students working individually, as noted in previous research (Kirschner, Paas, & Kirschner, 2009; Laughlin, Bonner, & Miner, 2006). While there were variations for which treatment condition did better on a specific outcome measure, on the whole the use of a group activity was more beneficial for outcome measures.

8. General discussion

Studies 1 and 2 required students to highlight and annotate an article. Within the framework of SAM-LS, there were variations in how the students were asked to use it. For this paper we looked at variations in highlighting, annotating, model comparison and reflection, along with the variation of individual or group tasks. Study 1 showed that the use of SAM-LS compared to a control condition did now show statistically significant differences on reading comprehension, critical thinking, and meta-cognition for individual students given the particular setting for Study 1. For the four different SAM-LS strategies, there were no significant differences. With respect to the change in skills over the course, there was significance in critical-thinking skills over the course when we combined the treatment effects with the control condition. There was no such change for reading comprehension and meta-cognition.

Whereas Study 1 did not consider a group effect, Study 2 did compare individuals and groups. For Study 2, we did not vary the instructional strategy across treatment groups, but had participants either complete the learning task as an individual or as a team. In this setting, the participants that worked in teams for either the first part or second part of the instructional task had greater learning, statistically significant, for reading comprehension and meta-cognition. We did not see a significant group effect for critical-thinking skills. In this study we did not compare the treatments with a control condition.

Based on these two studies, it seems that the annotation, highlighting, and reflection process does not have a strong impact if carried out by an individual. An individual could learn as much working with a textbook and not using the CSCL framework as presented in SAM-LS. However, if a student can team up with another student or two, there is a definite effect in learning of basic comprehension and meta-cognition as compared to students who work with the tool alone.

The design of SAM-LS was based on a theoretical framework comprised of social annotation, instructional design principles, team-based learning, and CSCL. Except for instructional design principles, the other components are linked to multiple persons interacting for the purpose of learning. The results of the two studies confirm two things: one, that a technology even implemented with an instructional strategy may or may not be better than a non-technology solution; two, there seems to be a difference when using the same instructional strategy when implemented for an individual as compared to a team.

Study 1 essentially embodied the implementation of the instructional design principles along with non-social annotations and omitting the team-based learning as well as the CSCL functionality. It seems that the SAM-LS without the "social" component is no more effective than having students read a book. When we added the "social" component back for three of the four treatment conditions (IA-GC, GA-IC, GA-GC), we did see a significant effect as compared to the IA-IC treatment. The group effects on reading comprehension would be expected based on the Lavagnino (1997) argument that annotations would clarify and fix reactions to text. Since this is believed to help the person's process of interpretation, the resultant effect may have been the higher scores on meta-cognition. Also, annotation supports increased memory (Anderson & Armbruster, 1984; Bradley & Vetch, 2007). This effect was manifested in the group condition, but the effect was not seen individually. There may have been a team effect where the annotation was critical in getting the students to think about a problem, but the team interaction may have been the catalyst encoding the ideas, thereby facilitating recall of the reading.

From the team-based learning framework, the interaction and engagement as described by Lightner et al. (2007) would lead to the greater learning (Clark et al., 2008; Haberyan, 2007; Thompson et al., 2007). Although we did find greater learning with the teams in Study 2, research has not specified the types of learning that would be specific to team-based learning. Due to the increased engagement from learning in a team, the level of reading comprehension was significant. Perhaps the meta-cognition was more pronounced because team members are required to communicate more about their thinking when working as a team. We are not sure why the critical thinking was not different for the individual as compared to the teams, but a simple explanation could be that critical thinking might be more practiced in teams where a high level of trust is present. We did not measure team trust, but trust is typically developed over time, and our teams simply did not have the elapsed time that may have been needed for high trust levels needed for more substantial interactions.

For Study 2, working in teams might have interaction benefits in having multiple learners work together at the same location using

SAM-LS strategies. As we have seen, there were stronger learning measures when students worked on assignments in small groups rather than individually. The big difference between the individual treatments and the group treatments was the ability of the group (who were physically in the same location during the activities) to talk about and share ideas. These effects support the notion that having text annotations as well as some form of verbal explanations (peer talk) has a significant effect on student learning, specifically in reading comprehension and meta-cognition. The ability to communicate with one's teammates affords the teammates different advantages compared with virtual teams or individuals who do not have anyone with whom to communicate.

Although we worked hard to ensure compliance with the intervention conditions, neither Study 1 nor Study 2 directly observed students' performance on the highlighting and annotating activity. It is plausible that students could have made poor choices in deciding what information to highlight, and subsequently performed poorly on the comparison activity. If this were the case, due to the random assignment of instructional interventions, this effect would have been equally distributed throughout the various instructional methods. Even with this case, the results from Study 1 indicate that the SAM-LS treatments did not have a different effect on the outcome measures than the regular-conventional instruction.

The next steps in this research would be to conduct additional studies to advance the current findings. Additional study designs would need to observe instructional conditions over prolonged periods of time, as well as considering various process measures to determine the individual effects of learners' skill level. In addition to extended experimental research, a design-based research methodology might produce meaningful results related to the effects of the student learning process. In design-based research, one can measure the intermediate or process outcomes (Johnson & O' Connor, 2008; Sandoval, 2004). Within the context of the SAM-LS approach, intermediate outcomes would include students' performance on the *highlight and annotate*, as well as team interaction measures. This could include the evaluation and interpretation of students' annotation activities, thereby revealing the sources and characteristics of comprehension failures. This information could help focus on the selection of strategies for overcoming misconceptions, and provide a basis for monitoring and evaluating changes in student performance. Such intermediate measures may have a strong effect on the current outcome measures (e.g., students improve reading comprehension, critical thinking, and related meta-cognitive skills). The findings from these process measures could provide insights that would drive the improvement of the SAM-LS strategies to enable better use of the technology tools. In future studies of the SAM-LS approach, design-based research will provide a flexible, iterative process for "engineering" an effective alternative to remedial courses and programs.

Although future studies are needed to improve our understanding of the effect of the SAM-LS, current results suggest that there are potentially positive benefits in using SAM-LS in student teams in the regular first-year college curriculum as an instructional strategy to facilitate basic reading comprehension, critical thinking, and meta-cognitive skills.

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References

- ACT (2005). Average national ACT score unchanged in 2005; Students graduate from high school ready or not Retrieved February 18, 2008. Available from <http://www.act.org/news/releases/2005/8-17-05.html>.
- Adelman, C. (1996). The truth about remedial work: It's more complex than windy rhetoric and simple solutions suggest. *The Chronicle of Higher Education*, 43(6), 56.
- Adler, M. J., & Van Doren, C. (1972). *How to read a book*. New York: Simon & Shuster.
- Anderson, T. H., & Armbruster, B. B. (1984). Studying. In P. D. Pearson, R. Barr, M. Kamil, & P. Mosenthal (Eds.). *Handbook of reading research* (Vol. 1, pp. 657–679). New York: Longham.
- Boylan, H. R. (1999). Harvard symposium 2000: Developmental education: Demographics, outcomes, and activities. *Journal of Developmental Education*, 23, 2–8.
- Bradley, J., & Vetch, P. (2007). Supporting annotation as a scholarly tool-experiences from the online Chopin variorum edition. *Literary and Linguistic Computing*, 22(2), 225–241.
- Brown, A. L., & Campione, J. C. (1990). Communities of learning and thinking, or a context by any other name. In D. Kuhn (Ed.), *Developmental perspectives on teaching and learning thinking skills* (Vol. 21). Contributions to human development (pp. 108–126). Basel: Karger.
- Buneman, P., & Steedman, M. (2002). Annotation – The new medium of communication. In *Workshop on Grand challenges in computer science*. Edinburgh.
- Clark, M. C., Nguyen, H. T., Bray, C., & Levine, R. E. (2008). Team-based learning in an undergraduate nursing course. *Journal of Nursing Education*, 47(3), 111–117.
- Collazos, C. A., Guerro, L. A., & Pino, J. A. (2004). Computational design principles to support the monitoring of collaborative learning processes. *Advanced Technology for Learning*, 1(3), 174–180.
- Collins, A., Brown, J. S., & Newman, S. E. (1989). Cognitive apprenticeship: Teaching the craft of reading, writing and mathematics. In L. B. Resnick (Ed.), *Knowing, learning and instruction: Essays in honor of Robert Glaser* (pp. 453–494). Hillsdale, NJ: Lawrence Erlbaum.
- Cox, S. R., Freisner, D., & Khayum, M. (2003). Do reading skills courses help under-prepared readers achieve academic success in college. *Journal of College Reading and Learning*, 33(2), 170–196.
- Dillenbourg, P. (2002). Over-scripting CSCL: The risks of blending collaborative learning with instructional design. In P. A. Kirschner (Ed.), *Three worlds of CSCL: Can we support CSCL* (Vol. 61–91). Heerlen: Open Universiteit Nederland.
- Donahue, P. L., Voelkl, K. E., Campbell, J. R., & Mazzeo, J. (1999). *NAEP 1998 reading: Report card for the nation and the states* (No. NCES 1999–500). Washington, DC: National Center for Educational Statistics.
- Ericsson, K. A. (2002). Attaining excellence through deliberate practice: Insights from the study of expert performance. In M. Ferrari (Ed.), *The pursuit of excellence in education* (pp. 21–55). Hillsdale, NJ: Lawrence Erlbaum.
- Everson, H. T., & Tobias, S. (1998). The ability to estimate knowledge and performance in college: A metacognitive analysis. *Instructional Science*, 26, 65–79.
- Fisher, A. (2001). *Critical thinking: An introduction*. Cambridge: Cambridge University Press.
- Glover, I., Xub, Z., & Hardaker, G. (2007). Online annotation – Research and practices. *Computers and Education*, 49(4), 1308–1320.
- Gokhale, A. (1995). Collaborative learning enhances critical thinking. *Journal of Technology Education*. Available from <http://scholar.lib.vt.edu/ejournals/JTE/jte-v7n1/gokhalejte-v7n1.html>.
- Haberyan, A. (2007). Team-based learning in an Industrial/Organizational Psychology course. *North American Journal of Psychology*, 9(1), 143–152.
- Hartman, H. J. (2001). Developing students' meta-cognitive knowledge and skills. In H. J. Hartman (Ed.), *Meta-cognition in learning and instruction: Theory, research, and practice* (pp. 33–68). The Netherlands: Kluwer Academic Publishers.
- Hoffman, B., McCurden, M., Schraw, G., & Hartley, K. (2008). The effects of informational complexity and working memory on problem-solving efficiency. *Asia Pacific Education Review*, 9(4), 464–474.
- Johnson, D. W., & Johnson, H. (1991). *Learning together and alone: Cooperation, competition, and individualization* (3rd ed.). Englewood Cliffs, NJ: Prentice Hall.
- Johnson, T. E., & O' Connor, D. L. (2008). Assessment of group learning processes. In M. D. Merrill, J. van Merriënboer, J. M. Spector, & M. Driscoll (Eds.), *Handbook of research on educational communications and technology* (3rd ed., pp. 776–783). New York: Lawrence Erlbaum.
- Kirschner, F., Paas, F., & Kirschner, P. (2009). Individual and group-based learning from complex cognitive tasks: Effects on retention and transfer efficiency. *Computers in Human Behavior*, 25(2), 306–314.
- Koschmann, T., Hall, R., & Miyake, N. (Eds.). (2002). *CSCL 2: Carrying forward the conversation*. Mahwah, NJ: Lawrence Erlbaum.
- Kirschner, P., Strijbos, J. W., Kreijns, K., & Beers, P. J. (2004). Designing electronic collaborative learning environments. *Educational Technology, Research and Development*, 52(3), 47–66.
- Krathwohl, D. R. (2002). A revision of bloom's taxonomy: An overview. *Theory into Practice*, 4, 212–218.
- Kumpulainen, K., & Wray, D. (Eds.). (2002). *Classroom interaction and social learning*. London: RoutledgeFalmer.

- Lancaster, K., & Strand, C. (2001). Using the team-learning model in a managerial accounting class: An experiment in cooperative learning. *Issues in Accounting Education*, 16(4), 550–567.
- Laughlin, P. R., Bonner, B. L., & Miner, A. G. (2006). Groups perform better than the best individuals on letters-to-numbers problems: Effects of group size. *Journal of Personality and Social Psychology*, 90(4), 644–651.
- Lavagnino, J. (1997). Reading, scholarship, and hypertext editions. *The Journal of Electronic Publishing*, 3(1). Available from <http://www.stg.brown.edu/resources/stg/monographs/rshe.html>.
- Lebow, D. G., Lick, D. W., & Hartman, H. J. (2004). HyLighter and interactive annotation: New technology to develop higher-order thinking skills. *Inquiry: Critical Thinking Across the Curriculum*, XXIII(1&2), 69–79.
- Lehtinen, E. (2003). Computer-supported collaborative learning: An approach to powerful learning environments. In E. DeCorte, L. Verschaffel, N. Entwistle, & J. van Merriënboer (Eds.), *Unravelling basic components and dimensions of powerful learning environments* (pp. 35–53). Amsterdam: Elsevier.
- Lockhead, J., & Clement, J. (Eds.). (1979). *Cognitive process instruction – Research on teaching thinking skills*. Philadelphia: Franklin Institute Press.
- Lightner, S., Bober, M. J., & Willi, C. (2007). Team-based activities to promote engaged learning. *College Teaching*, 55(1), 5–18.
- McCabe, R. H. (2000). *No one to waste: A report to public decision-makers and community college leaders*. Washington, DC: Community College Press.
- Mendelman, L. (2007). Critical thinking and reading. *Journal of Adolescent & Adult Literacy*, 51(4), 300–302.
- Merrill, M. D. (2002). First principles of instruction. *Educational Technology, Research and Development*, 50(3), 43–59.
- Michaelsen, L. K., Knight, A. B., & Fink, L. D. (2004). *Team-based learning*. Sterling, VA: Stylus.
- Moss, D. M. (2000). Bringing together technology and students: Examining the use of technology in a project-based class. *Journal of Educational Computing Research*, 22(2), 155–169.
- Oudenhoven, B. (2002). Remediation at the community college: Pressing issues, uncertain solutions. *New Directions for Community Colleges*, 117, 35–44.
- Peterson, S. E. (1992). The cognitive functions of underlining as a study technique. *Reading Research and Instruction*, 31(2), 49–56.
- Porter-O'Donnell, C. (2004). Beyond the yellow highlighter: Teaching annotation skills to improve reading comprehension. *English Journal*, 93(5), 82–89.
- Sandoval, W. A. (2004). Developing learning theory by refining conjectures embodied in educational designs. *Educational Psychologist*, 39(4), 213–223.
- Schellings, G. L. M., & van Hout-Wolters, B. H. A. M. (1995). Main points in an instructional text, as identified by students and by their teachers. *Reading Research Quarterly*, 30, 742–756.
- Simpson, M. L., & Nist, S. L. (1990). Textbook annotation: An effective and efficient study strategy for college students. *Journal of Reading*(34), 122–131.
- Srinivas, H. (2008). Collaborative learning enhances critical thinking. Available from <http://www.gdrc.org/kmgmt/c-learn/> Accessed 15.04.10.
- Sternberg, R. J. (1986). *Intelligence applied: Understanding and increasing your intellectual skills*. San Diego, CA: Harcourt Brace Jovanovich.
- Thompson, B., Schneider, V., Haidet, P., Levine, R., McMahon, K., Perkowski, L., et al. (2007). Team-based learning at ten medical schools: Two years later. *Medical Education*, 41(3), 250–257.
- van Merriënboer, J. J. G. (1997). *Training complex cognitive skills: A four-component instructional design model for technical training*. Englewood Cliffs, NJ: Educational Technology Publications.
- Wahlstrom, B., & Scruton, C. (1997). Constructing texts/understanding texts: Lessons from antiquity and the Middle Ages. *Computers and Composition*, 14(3), 311–328.
- Wirt, J., Choy, S., Gerald, D., Provasnik, S., Rooney, P., Watanabe, S., et al. (2002). *The condition of education 2002 (No. NCES 2002-025)*. Washington, DC: National Center for Educational Statistics.
- Wolfe, J. L. (2002). Annotation technologies: A software and research review. *Computers and Composition*, 19, 471–497.